

Outline of the Report

Assignment Task: Each student must select **one topic from the syllabus**, explore its **real-world applications**, and demonstrate its **implementation** (analytical, computational, or experimental).

Structure of the Report:

1. **Title Page** Title of the topic, name, roll number, course, and date.
2. **Introduction** Brief overview of the chosen topic and its theoretical background.
3. **Motivation** Why this topic was chosen and its relevance to real-world problems.
4. **Literature Review** Summary of existing works, studies, or applications related to the topic.
5. **Problem Statement & Objectives** Define the specific problem addressed and outline the main objectives.
6. **Data Set (if applicable)** Description of data collected or used for analysis or implementation.
7. **Methodology** Explain the methods, models, or algorithms used for the study or implementation.
8. **Implementation / Work Done** Details of your analysis, computations, or results obtained.
9. **Results and Discussion** Present findings, interpretations, and comparisons with existing work.
10. **Conclusion and Future Scope** Summarize key insights and suggest possible extensions or improvements.
11. **References** List all sources referred to (books, papers, websites).